Ioannis Mouratidis

Austin, Texas

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Experienced research engineer looking to advance AI safety and evals. 38 publications (12 first/senior author) and 3 patents in large-scale data analysis and ML applications in biology and AIxBio safety. Co-founded startup in cancer diagnostics, authored grants securing $4M+ in funding and currently focus on AI safety and alignment research.

Experience

Timaeus Research — **Berkeley, CA**
Research Engineer — 05/2026 – 09/2026 (contract)

- Contributing to AI safety research at a leading alignment-focused organization, studying how capabilities and values emerge during model training, with the goal of building scalable interventions to align frontier models.
- Developing core research infrastructure and translating exploratory prototypes into robust, scalable systems.

University of Texas at Austin — **Austin, TX**
Research Engineer/Scientist Associate V (senior) — 08/2025 – 05/2026

- Led a 5-member team and directed 12 research projects (4 independently, 8 collaboratively) at the intersection of AI safety and biology, developing novel methods to evaluate and restrict dual-use capabilities of agentic LLMs and biological foundation models, including misuse potential, adversarial robustness, and synthetic biology risk.
- Spearheaded red-teaming of agentic AI scientists discovering novel vulnerabilities.
- Evaluated adversarial robustness of genomic foundation models, including susceptibility to data poisoning.
- Directed benchmarking of genomic language models against real genomes, evaluating capabilities in synthetic biology and identifying systematic discrepancies that can be used to identify synthetically generated sequences.
- Developed tamper-resistant weight-locking methods for open-weight biological AI models.

The Pennsylvania State University — **Hershey, PA**
Research Associate & Research Assistant — 08/2023 – 08/2025

- Built scalable High-Performance Computing (HPC) pipeline in Slurm enabling the identification of pathogen biomarkers across 45,000 genome assemblies; led scaling to 500,000 assemblies.
- Developed the *kmerDB* database, consolidating information on short DNA and protein sequences across over 60K genomes and proteomes; co-developed 4 additional public scientific databases.
- Created open-source *Zseeker* Python tool; co-developed 5 additional open-source bioinformatics tools.
- Delivered oral presentations at 4+ national and international conferences, ranging from 150-300 attendees.
- Supervised 8+ researchers and coordinated collaborative projects across multiple research groups; mentored 5 junior researchers with no prior bioinformatics experience to their first first-author publications.
- Designed research projects and drafted grant applications resulting in $3M+ in competitive funding.

Neomer Diagnostics — **Brussels, Belgium**
Co-founder and Chief Technical Officer — 01/2022 – 05/2023

- Co-founded diagnostics startup and coordinated across technical, regulatory, and investor stakeholders to translate patented nullomer research into clinical platform.
- Developed ML pipeline for cancer detection from liquid biopsies, achieving AUC ranging from 0.89 to 0.94.
- • Contributed to securing $850K through technology development and investor demos.

Independent Contractor — **Brussels, Belgium**
Software Engineering Consultant — 02/2020 – 08/2023

- Optimized ML pipeline using Julia; 50x speedup, saving >3 months of computational time and $50K+ in costs.
- Deployed and managed genomic analysis pipeline to GCP, enabling processing of 14 TB of cfDNA sequencing data.

European Commission — **Brussels, Belgium**
Administrative Agent — 04/2019 – 05/2021

- Provided operational and policy support while pursuing M.Sc. in Artificial Intelligence (2020-2021).

Education

The Pennsylvania State University **Hershey, PA**
Ph.D. in Bioinformatics (Cumulative GPA: 4.0) 08/2023 – 04/2026
Doctoral thesis: Computational and machine learning methods for k-mer sequence analysis in large-scale omics data

KU Leuven **Leuven, Belgium**
M.Sc. in Artificial Intelligence (Honours: Cum Laude) 09/2020 – 07/2023
Master's thesis: Frequentmers – a k-mer based approach for the detection of liver cirrhosis gut microbiome from mNGS

Aristotle University of Thessaloniki **Thessaloniki, Greece**
B.Sc. in Mathematics (Honours: Very Good) 10/2013 – 03/2018

Skills

Technical skills:

- Programming Languages: Python (Pandas, NumPy, scikit-learn, PyTorch, Matplotlib, FastMCP), Bash, Julia
- Tools & Platforms: High-Performance Computing (Slurm, Skypilot), Google Cloud Platform, Git, Docker
- Computational Methods: Empirical ML research, algorithm design and optimization, large-scale data analysis, machine learning pipelines, genomic language models
- AI Safety: AI evals, singular learning theory, dual-use risk assessment, red-teaming, threat modeling
- Research Skills: Study design, scientific writing, grant writing

Languages: English (Fluent), French (Fluent), German (Fluent), Japanese (Elementary), Greek (Native)

Honors and Awards

- Alumni Graduate School Association Dissertation Award, Penn State, 2026 (2% acceptance rate)
- Alumni Society Award, Penn State College of Medicine, 2024 (5% acceptance rate)
- University Graduate Fellow & Huck Distinguished Fellow, Penn State, 2023 (1% acceptance rate)

Patents

- Georgakopoulos-Soares, I., **Mouratidis, I.**, Provatas, K. (2025). System and Method for Extracting Neomers. U.S. Provisional Patent.
- Ahituv, N., Yizhar-Barnea, O., Georgakopoulos-Soares, I., **Mouratidis, I.**, Hemberg, M. (2023). Systems for mutation caller and methods of using the same. WO2024103003A3.
- Ahituv, N., Yizhar-Barnea, O., Georgakopoulos-Soares, I., **Mouratidis, I.**, Hemberg, M. (2022). Compositions comprising nullomers and methods of using the same for cancer detection and diagnosis. WO2022235718A3.

Selected Publications

1. Provatas, K., Self, A., **Mouratidis, I.**[+], & Georgakopoulos-Soares, I.[+] (2026). BioVeil MATRIX: Uncovering and categorizing vulnerabilities of agentic biological AI scientists. *arXiv*.
2. Koilakos, C., **Mouratidis, I.**[+], & Georgakopoulos-Soares, I.[+] (2026). Poisoning the genome: Targeted backdoor attacks on DNA foundation models. In *arXiv*.
3. Tzanakakis, A., **Mouratidis, I.**[+], & Georgakopoulos-Soares, I.[+] (2026). Fundamental limitations of genomic language models for realistic sequence generation. *bioRxiv*.
4. Georgakopoulos-Soares, I.*, Yizhar-Barnea, O.*, **Mouratidis, I.***, et al. (2025). Leveraging sequences missing from the human genome to diagnose cancer. *Communications Medicine*, *5*(1).
5. Wang, G.*, **Mouratidis, I.***, Provatas, K., et al. (2025). ZSeeker: an optimized algorithm for Z-DNA detection in genomic sequences. *Briefings in Bioinformatics*, 26(3).

Complete list of published work can be found in Google Scholar.

* Co-first author [+] Co-corresponding (supervising) author